How Low Can You Go? Lower Limits of Recoverable SMBH Mass in Virgo Cluster UCDs from JWST NIRSpec IFU Data

A. Lapeer [(1)], M. Valluri [(1)], E. Vasiliev [(2)], B. Tahmasebzadeh [(1)], M. Taylor [(3)]

[1]University of Michigan, Department of Astronomy, 1085 S. University Ave, Ann Arbor, MI 48109, USA

[2]University of Cambridge, Institute of Astronomy, Madingley Road, Cambridge, CB3 0HA, UK

[3]University of Calgary, 2500 University Drive NW, Alberta T2N 1N4, CA

M | LSA ASTRONOMY UNIVERSITY OF MICHIGAN

Abstract

Ultra-compact dwarf galaxies (UCDs), characterized by their abnormally dense stellar populations and ultra-compact nature, provide a unique tool in the study of galaxy and galaxy group evolution, and possibly exist as a link between various classes of stellar systems (e.g. globular clusters, dwarf ellipticals). Central black holes in such systems not only provide a useful asset in probing the nature of UCDs, but also in the constraint of the black hole mass function in general.

This investigation aims to probe the lower limit of recoverable central black hole mass in such systems via self-consistent orbit-superposition (Schwarzschild) modeling (Schwarszschild 1979). We employ the FORSTAND code (Vasiliev & Valluri 2020), an iterative, observationally-constrained Schwarszchild modeling routine, to recover black hole mass from mock kinematic data generated from static, self-consistent N-body representations of simplified UCD-like systems. The mock data are designed to mimic the spatial and spectral resolution of JWST NIRSpec IFU data. This analysis of mock data will enable us to interpret and model IFU kinematics from a Cycle I JWST Guest-observer program. We identified a lower limit of recoverable mass of order ~$10^{6.5}$ $M_\odot$, however further investigation is needed to constrain this limit to the fullest extent.

Modeling Regime & Mock Data

Prior to modeling, a simple, DF-based N-body representation is generated according to target parameters (e.g. mass, BH mass, half-light radius). The addition of other galactic components (e.g. DM halo, stellar bulge) is also possible, however this investigation employs simple spheroid systems composed of only a stellar component and central BH. Utilizing the FORSTAND code, mock IFU kinematics are then generated based on NIRSpec IFU parameters and the N-body representation.

When generating mock data, the N-body representation is first projected via a multi-Gaussian expansion (MGE) fit and Voronoi binned. Following this, B-spline line-of-sight velocity distributions (LOSVDs) are computed and subsequently converted to Gauss-Hermite (GH) moments. These components (MGE parameterization, GH moments, Voronoi binning scheme, and LOSVDs) are then passed to FORSTAND, assembled as one dataset, and used for modeling.

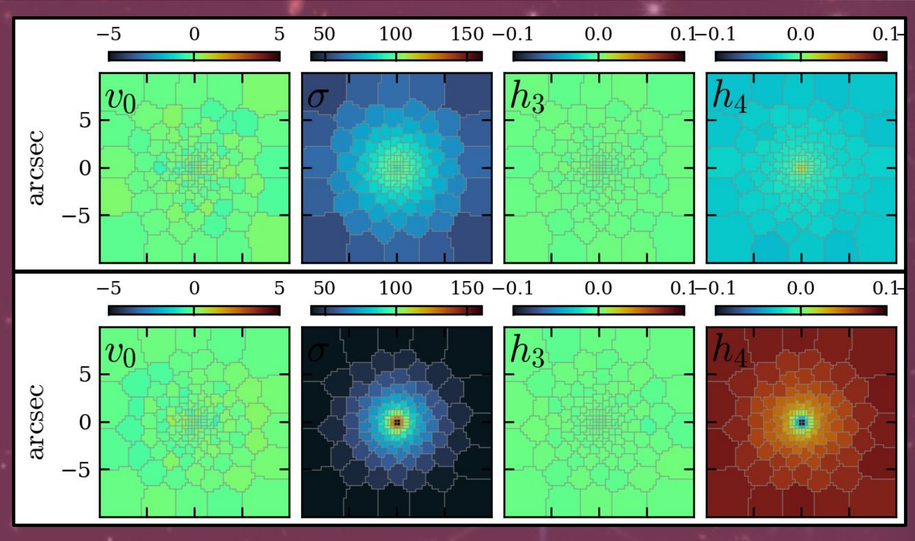


Figure 1: Mock kinematic data with $M_{BH} = 10^7$ M_* and $M_{Gal.} = 10^{9.5}$ M_* and varying degrees of anisotropy (determined by anisotropy coefficient β_0). Power law index of the inner density profile in both cases is $\gamma = 1$ and that of the outer density profile is $\beta = 4$. From left to right: velocity distribution, velocity dispersion, and GH moments. ***Top:*** Mock kinematics of a simple spheroidal stellar system with $\beta_0 = 0$. ***Bottom:*** Mock kinematics with identical parameters but $\beta_0 = 0.7$.

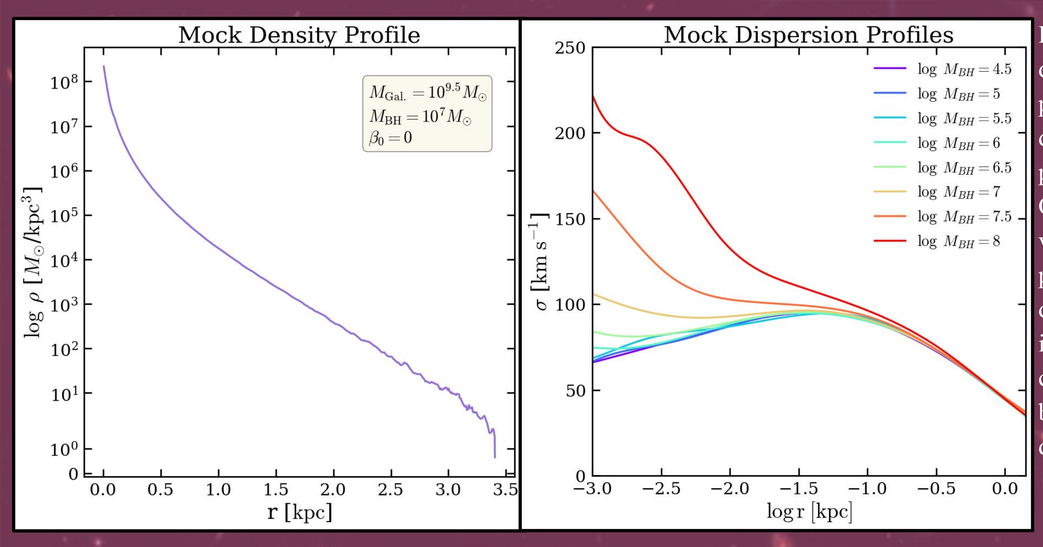


Figure 2: ***Left:*** One dimensional density profile of the mock data seen in the top panel of Fig (1). ***Right:*** One dimensional velocity dispersion profiles of each mock dataset used in this investigation. A clear correlation can be seen between peak velocity dispersion and M_{BH}.

A Brief Summary of the FORSTAND Modeling Regime

1. A three-dimensional density profile and discretization scheme is constructed from MGE parameterization.
2. Aperture boundaries are created from the Voronoi binning scheme.
3. Kinematic data is generated according to aperture boundaries from GH moments.
4. The above datasets (kinematics and density profile) are compiled together to form the target dataset.
5. A value (or range of values) is selected for M_{BH} (this parameter is invariant, each model is run for one value of M_{BH}).
6. Dark matter halo is generated based on initial mock parameters (in our case, no halo is generated).
7. From the target dataset, potential of the galaxy and its central BH are generated and combined (used for orbit integration).
8. A potential is created for a fiducial BH component (used to construct initial conditions which remain constant across models of varying M_{BH}).
9. Initial conditions are generated from the fiducial potential
10. Iterative modeling is run over a grid of M/L values while ensuring self-consistency via the Poisson equation, DF of the model, and CBE.

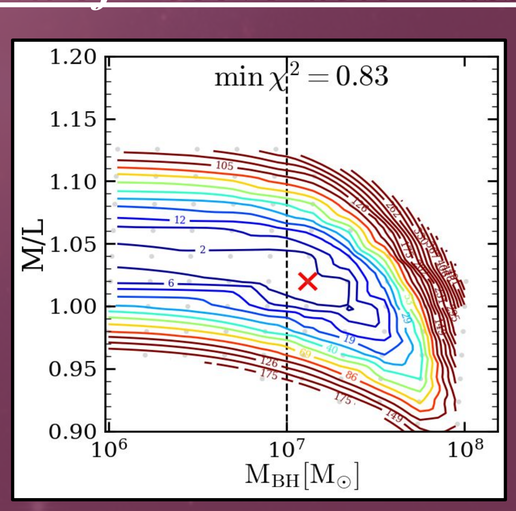
Models

M_{BH} ($M_\odot$)	M_{BH} / M_{Gal}	Best Fit M_{BH}	Upper Limit M_{BH}	Recovered M/L
$10^{4.5}$	0.001%	$10^{4.50}$	$10^{7.00}$	1.02
10^{5}	0.00316%	$10^{5.08}$	$10^{7.00}$	1.02
$10^{5.5}$	0.01%	$10^{5.08}$	$10^{7.00}$	1.02
10^{6}	0.0316%	$10^{5.50}$	$10^{7.10}$	1.02
$10^{6.5}$	0.1%	$10^{6.29}$	$10^{7.15}$	1.02
10^{7}	0.316%	$10^{7.11}$	$10^{7.38}$	1.02
$10^{7.5}$	1%	$10^{7.44}$	$10^{7.50}$	1.02
10^{8}	3.16%	$10^{7.83}$	$10^{8.00}$	1.061

Table 1: Above is a brief description of models used in this investigation. Parameters not listed above (anisotropy coefficient, host mass, scale radius, etc.) remained constant across all models and are available upon request. Fitting for each model was run over a linearly spaced range of 25 M_{BH} values from 4.5 to 10 solar masses. True M/L for each model is 1.0 due to our mock systems being composed of only a stellar component.

Identifying a Lower Limit of Recoverable Mass

Figure 3: An example of the chi-squared plane for a model with $M_{BH} = 10^7$ $M_\odot$. The best-fit model is designated by a red "X" while the true value of M_{BH} of the model is plotted as a dashed line. Note that this example was not run over the same array of M_{BH} values as the above models for the sake of visual clarity, and serves only as an example of typical chi-square plane contours used in this investigation.



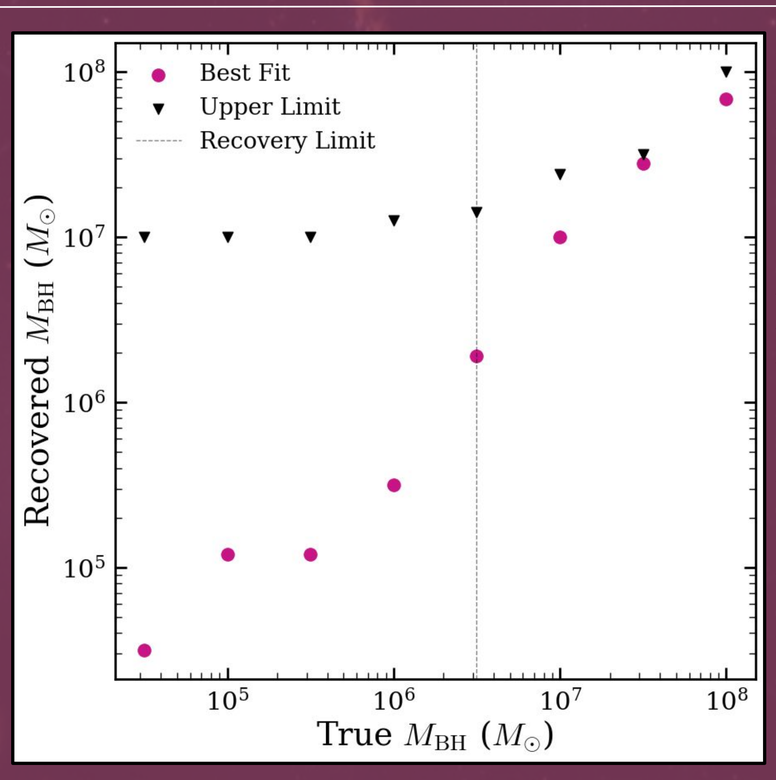


Figure 4: True M_{BH} plotted against best-fit M_{BH} for each model, along with the upper limit of M_{BH} for each model. A clear cutoff point can be seen at $M_{BH} = 10^{6.5}$ $M_\odot$ at which the upper limit of recoverable black hole becomes constant for all models below $10^{6.5}$ $M_\odot$. While the best-fit M_{BH} remains a relatively good indication of true M_{BH}, such a limit poses issues in the recovery of such values from real observations. Thus, further investigation is needed into this topic to gain a full understanding of the limitations of applying FORSTAND to UCD observations obtained from the JWST NIRSpec IFU.

References

Vasiliev, E., & Valluri, M. 2020, ApJ, 889, 39

Schwarzschild, M. 1979, ApJ, 232, 236

Vasiliev E., 2019, MNRAS, 482, 1525

Liu C., et al., 2020, ApJ, 250, 17